

Tumor control, eye preservation, and visual outcomes of ruthenium plaque brachytherapy for choroidal melanoma

Daniel Grossi Marconi^{1,*}, Douglas Guedes de Castro¹, Lievin Matos Rebouças¹, Gabriel Oliveira Bernardes Gil¹, Ricardo Cesar Fogaroli¹, Maria Aparecida Conte Maia¹, Maria Leticia Gobo Silva¹, Antonio Cassio Assis Pellizzon¹, Maria Marta Motono Chojniak²

¹Department of Radiation Oncology, AC Camargo Hospital, São Paulo/SP, Brazil

²Department of Ophthalmology, AC Camargo Hospital, São Paulo/SP, Brazil

ABSTRACT

PURPOSE: To evaluate outcomes in patients with posterior choroidal melanoma treated with ruthenium (¹⁰⁶Ru) brachytherapy.

METHODS AND MATERIALS: A retrospective single institutional analysis of 83 of 94 consecutive patients who underwent ¹⁰⁶Ru brachytherapy was performed. Disease was mainly staged as small- and medium-sized nonmetastatic melanoma. The main parameters evaluated were tumor control (local control [LC] and progression-free survival [PFS]) and ocular preservation (enucleation-free survival [EFS]). Besides, functional evaluation was performed and complications were described.

RESULTS: The median follow-up was 39 (6–83) months. The median values of height and maximal basal diameter were 4.3 and 9.3 mm, respectively. Median apical and basal doses were 100 and 307 Gy, respectively. The actuarial 2-year LC, PFS, and EFS were 96.2%, 96.2%, and 95.5%, respectively. Actuarial 5-year LC, PFS, and EFS were 93.6%, 93.6%, and 84.1%, respectively. Preinsertion visual acuity (VA) maintenance was 34% (equal or better than before treatment). Approximately 56% of patients stayed with a minimum functional VA of 0.1 or more, from whom more than half stayed with 0.5 or more. Cataract was seen in 16% of treated eyes, and glaucoma was the rarest complication, with an incidence of 3%.

CONCLUSIONS: Small- and medium-sized choroidal melanomas can be adequately treated with ¹⁰⁶Ru brachytherapy, with high rates of tumor control and ocular preservation. Moreover, acceptable incidence of complications such as glaucoma and cataract are seen, and a reasonable part of patients stay with a minimum functional VA. © 2013 American Brachytherapy Society. Published by Elsevier Inc. All rights reserved.

Keywords:

Choroidal melanoma; Uveal neoplasms; Brachytherapy; Ruthenium-106

Introduction

Uveal melanoma is the most common primary neoplasm of the eye with an annual incidence of 4.3 per million and

was historically treated with enucleation. Aiming at improving cosmetics and quality of life without compromising survival, alternative eye-sparing treatments such as brachytherapy, charged particle radiotherapy, stereotactic radiotherapy, local resection, and phototherapy were developed (1).

Several studies have shown that these conservative procedures do not increase the risk of distant metastasis and death (2, 3) and are reasonable alternatives to enucleation. The biggest one to assess the efficacy of brachytherapy was the Collaborative Ocular Melanoma Study (COMS) that included 1317 patients mainly with medium-sized choroidal melanoma and compared iodine (¹²⁵I) plaque to enucleation. There was no difference between these two treatments, with

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Conflicts of interest: None.

* Corresponding author. Department of Radiation Oncology, AC Camargo Hospital, Antonio prudente, 211, Liberdade, São Paulo/SP, Brazil 01504-000. Tel.: +55-11-2189-5000; fax: +55-17-2189-5104.

E-mail address: dgmarconi@gmail.com (D.G. Marconi).

comparable 5-year survival (81 vs. 82%) and 5-year death rates from distant metastasis (11% vs. 9%) (4).

The main purpose of brachytherapy seems to be proven after that (comparable cure). At present, the objectives are maintenance of vision and improvement of cosmetics and quality of life. Many isotopes were tested and since it was introduced, ^{106}Ru has been increasingly used.

In this context, it was decided to perform a retrospective review of our results with ^{106}Ru brachytherapy for small- and medium-sized posterior choroidal melanomas. The main parameters evaluated were tumor control (local control [LC] and progression-free survival [PFS]) and ocular preservation (enucleation-free survival [EFS]). Besides, functional evaluation was performed, and complications such as cataracts, visual acuity (VA), and glaucoma were described.

Methods and materials

The records from 94 consecutive patients with the presumed diagnosis of choroidal melanoma who underwent brachytherapy using ^{106}Ru between January 2004 and September 2008 at AC Camargo Hospital in São Paulo, Brazil were reviewed.

Inclusion/exclusion criteria

All patients were aged 18 or older at the time of implant and all of them signed an informed consent describing the possible toxicities from treatment and allowing future studies like this.

Patients with follow-up shorter than 6 months, those whose lesions were located at the anterior compartment, and/or those presenting metastases at the time of diagnosis were excluded. This study had ethics approval before initiation.

Technical parameters

Ruthenium applicators (CGD model—22.3 mm diameter, 6.1 mm height, and 13 mm radius) manufactured by Bebig (Eckert & Ziegler BEBIG GmbH, Berlin, Germany) were used.

The choice of the plaque diameter was made by adding 2–4 mm to the biggest tumor diameter to have a safety margin of at least 1–2 mm on each side.

A 100 Gy dose delivered by a minimum 0.6 Gy/h dose rate was prescribed at the apex for any lesion larger than 5 mm. Smaller lesions had doses delivered at 5 mm from their base. One millimeter corresponding to sclera thickness was added to the true height value. Dose to the base was then calculated (4).

Tumor size was classified according to the COMS criteria (5). The small tumors measured less than 2.5 mm in apical height and 5 mm or less in longest basal diameter, the medium tumors ranged from 2.5 to 10 mm in apical

height and 16 mm or less in the longest basal diameter. The tumors classified as large had a longer basal diameter of more than 16 mm and height of more than 2 mm or an apical height of more than 10 mm.

The plaque was applied to the patient under general anesthesia, and the lesion localization was performed by direct visualization, transillumination, or indirect ophthalmologic examination. Patients remained hospitalized in appropriate rooms during the time of irradiation, and the plaque removal was made under general anesthesia as well.

Outcome measures

Tumor response was evaluated by fundoscopy and ultrasound. Local recurrence was defined by the presence of signals of tumor activity or documented growth.

Metastases were investigated routinely and in symptomatic patients. Disease progression was defined by any local or distant signal of tumor activity.

Appearance of cataract and glaucoma was evaluated based on citation in patient record. A measured eye pressure higher than 21 mm Hg defined glaucoma. Enucleation was performed owing to local recurrence or late toxicities, and these parameters were separately analyzed.

The main parameters evaluated were tumor control (LC and PFS) and ocular preservation (EFS). They were calculated using Kaplan–Meier survival curves. Besides, functional evaluation was performed, and complications such as cataracts, loss of VA, and glaucoma were described.

Results

Follow-up

Exclusion rules resulted in retaining only 83 of 94 patients who underwent ^{106}Ru plaque during this period. The main cause of exclusion was follow-up duration of less than 6 months (many patients did the follow-up at their original hospital). It resulted in a median follow-up of 39 (6–83) months.

Patients and tumor characteristics

Most of the patients were female, corresponding to 60% of all cases. From the total, 58% of the lesions occurred in the left eye. The median age was 60 (19–84) years, and comorbidities were not evaluated. Median treatment time was 3 days (minimum 2 and maximum 12 days). Tumor and treatment features are listed in Table 1.

Tumor control

There were eight enucleations—4 owing to tumor progression and four owing to toxicity—and only one distant failure (liver, lung, and central nervous system [CNS] metastasis after 72 months of treatment). There was no combined failure.

Table 1
Tumor and treatment features

Treatment features	Median value (range)
Height (mm)	4.3 (2.3–7)
Basal diameter (mm)	9.3 (3.8–15.8)
Apical dose (Gy)	100 (43.3–206.9)
Basal dose (Gy)	307 (251–420)
Treatment time (h)	72 (48–288)

The actuarial 2- and 5-year LC, PFS, and EFS are described in Table 2. Actuarial Kaplan–Meier curves are described in Figs. 1–3.

Toxicity

The four enucleations occasioned by toxicities were: two vitreous bleeding (impossibility to keep lesion growth follow-up), one intractable painful blind eye, and one posterior uveal necrosis. VA maintenance was 34% (equal or better than before the plaque insertion). Approximately 30% kept VA of 0.5 or more and approximately 56% stayed with a minimum functional vision of 0.1 or more.

In relation to other toxicities, cataract was seen in 16% of patients and glaucoma had an incidence of 3%. There was no case of actinic neuropathy.

Discussion

This cohort consists of a consecutive series of patients evaluated in a historical follow-up design. Parameters evaluation was performed by a single-variable analysis only, as the material was too small for multivariate analysis and could have potential confounding factors. Each one is discussed separately.

Tumor control

Increased risk of local recurrence has been found due to low radiation dose to tumor apex (6–8) and low dose rate to the apex (8). The COMS dose prescription is widely used as a standard for eye plaque treatment. Earlier the prescription dose was 85 Gy to the tumor apex, American Brachytherapy Society has recommended the same, with a dose rate of 0.60–1.05 Gy/h, using the COMS dosimetry assumptions and plaque construction techniques (5). Thus, we consider it adequate if the median dose of 100 Gy is achieved, and although the dose rate was not available, a minimum of 0.60 Gy/h is fulfilled in our department.

Table 2
Tumor control and ocular preservation

Characteristics	2-y survival (%)	5-y survival (%)
Local control	96.2	93.6
Progression-free survival	96.2	93.6
Enucleation-free survival	95.5	84.1

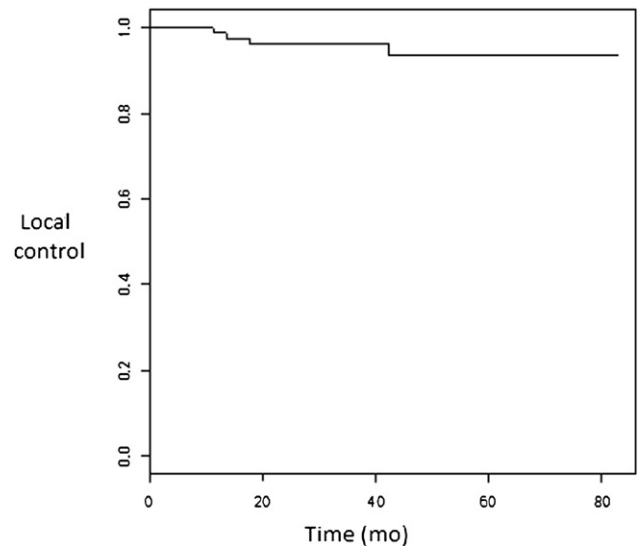


Fig. 1. Local control.

With this dosimetry calculation, the COMS demonstrated excellent LC rates and survival outcomes. Using ^{125}I , at 5 years, they reported local failure and disease-free survival rates of 10% and 90%, respectively. By comparison, our results demonstrated a local failure and survival rate of 8% and 92%, respectively. Additionally, the COMS demonstrated excellent ocular preservation with a 5-year enucleation rate of only 13%, a little better than our 15.9% (seven enucleations at 5 years).

Tumor location could be a confounding factor when depending on the position, as it would increase the risk of local recurrence—when close to the optic disc or fovea, it might pose as a problem in plaque positioning. In this series, we did not exclude those five high-risk patients, which would result in compromising results when

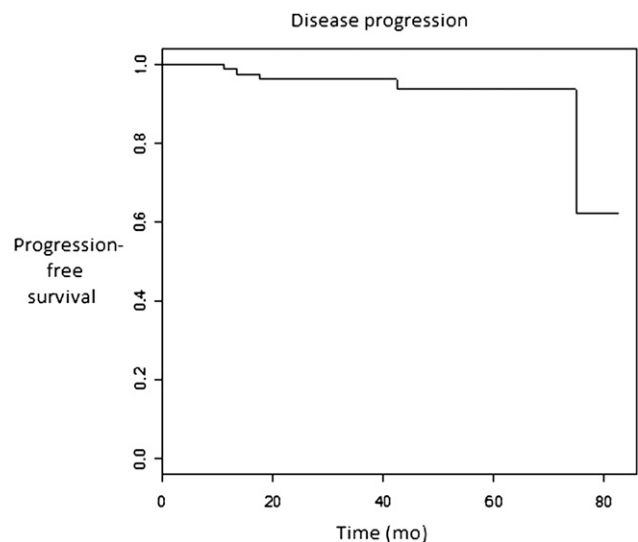


Fig. 2. Progression-free survival.

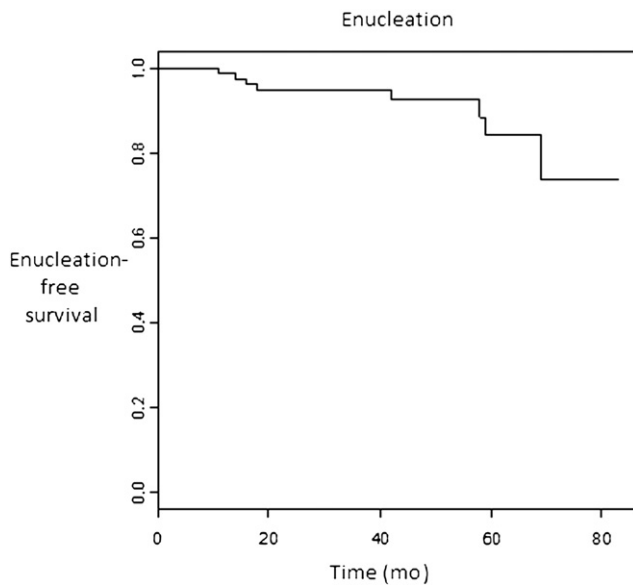


Fig. 3. Enucleation-free survival.

compared with other series, such as COMS (that historically excluded peripapillary tumors). Despite this, the 93.6% 5-year LC achieved was considered optimal when compared with 73% (9), 59% (10), and 78% (11) in other ruthenium series, and even comparable with results of peripapillary tumors treated with ^{125}I and heavy-charged particles (12). Other ruthenium series are presented in Table 3.

In fact, the relatively short follow-up makes us expect some late recurrences, as previously demonstrated by Char *et al.* (18). In this series, among the 996 analyzed treatments, there were 11 local recurrences after a period of 5 years, curiously none affected by either high-risk clinical or dosimetric features (like low-dose rate or peripapillary tumor). The only distant failure (liver, lung, and CNS) occurred 72 months after implant, and the patient is still alive.

Enucleation

The main reason for enucleation after brachytherapy is local tumor recurrence, with an incidence of 51–75%

(8, 16, 19), the remaining owing to toxicity. In our series, from the eight enucleations, four were caused by local recurrence.

According to Bergman *et al.* (16), in whose series the cumulative risk of enucleation was 16.8% in 5 years, most (76.2%) of all enucleations expected in 10 years were performed within the first 3 years after treatment, 75% because of inadequate LC of tumor. Our study has a median follow-up exceeding 3 years, which allows us to adequately evaluate the enucleations performed. The 5-year probability for eye preservation achieved was 84.1%, and we consider it reasonable when compared with 72% (9), 82% (11), 83% (16), and 85% (10) from other series (see Table 2).

Vision

Vision loss has many causes, including nonradiation-related diseases, such as age-related macular degeneration or senile cataract, but it was not accounted for. The presented vision loss might therefore be an overestimation of radiation side effects.

A large Swedish ruthenium series showed a 5-year probability for retaining VA of 0.5 or better of 31% and better than 0.1 of 49% (16), and another large series of radiation brachytherapy (iodine in 59%, cobalt in 27%, iridium in 9%, and ruthenium in 5% of patients) showed a 5-year probability for retaining VA better than 0.1 of 66% (20).

In the COMS (^{125}I) study, the 3-year probability for retaining vision better than 0.1 was 57%. Here, the two poor vision outcomes used to evaluate VA outcome at follow-up examination included a VA of 0.1 or worse or loss of six lines or more of VA from baseline (4).

According to Damato *et al.* (14), the actuarial probability of conservation of 20/40 or better was 0.80 (95% confidence interval [CI]: 0.75–0.84) at 2 years, 0.67 (95% CI: 0.61–0.73) at 5 years, and 0.55 (95% CI: 0.45–0.65) at 9 years.

Our results showed that approximately 30% of patients had VA of 0.5 or more and approximately 56% stayed with a minimum functional vision of 0.1 or more. When comparing with other series, these numbers are in concordance.

Table 3
Other series

Study	Patients	Medium basal diameter (mm)	Medium height (mm)	Medium apical dose (Gy)	Medium basal dose (Gy)	LC (%)	EFS (%)	VA (%)
Verschueren <i>et al.</i> (13)	425	10.9 (4.8–15.9)	4.2 (1.2–9.3)	—	400–800	5-y: 96	5-y: 95.6	5-y: 52
Damato <i>et al.</i> (14, 15)	458	10.6 (5–16.6)	3.2 (0.7–7)	80–100	300–400	5-y: 97.9	—	5-y: 57
Bergman <i>et al.</i> (16)	579	10 (3–23)	4.3 (1–13.3)	—	—	3-y: 89.6	5-y: 81.7	5-y: 49
Dias <i>et al.</i> (17)	20	—	3.45 (1.14–6)	85	248	19-mo: 75	—	19-mo: 70
Isager <i>et al.</i> (9)	55	12 (6–21)	5.8 (1.8–11)	100	—	5-y: 73	5-y: 72	5-y: 40
Current series	83	9.3	4.3	100	307	5-y: 93.6	5-y: 84.1	5-y: 56

LC = local control; EFS = enucleation-free survival; VA = visual acuity.

The values of VA represent a minimum functional vision of > 0.1.

Other complications

Choroidal melanomas are relatively resistant to irradiation; thus, the brachytherapy doses needed to control these tumors are associated with a substantial risk of radiation damage to the retina and optic nerve. Larger tumors and more anterior ones are more likely to develop this complication after brachytherapy, once these parameters affect the radiation dose to the lens.

According to the COMS report number 27 (21), the median time to the first report of cataract was 2.5 years (range: 6 months–5 years). By 3 years, 54% of the studied eyes were reported to have a cataract (95% CI: 50–59%), and by 5 years, 83% (95% CI: 79–87%). Although this series did not evaluate tumor location in the eye, our general data are in accordance with international literature: a total of 16% developed this complication after median follow-up of more than 3 years. Obviously, this number tends to increase over time. Other complications, such as glaucoma, appeared with low frequency in this series, and they are in accordance with international literature (22).

Conclusion

Ruthenium brachytherapy can be regarded as an appropriate treatment option for small- and medium-sized choroidal melanomas once it has excellent rates of LC and eye preservation. Besides, acceptable indices of toxicities such as cataract and glaucoma were achieved, and a reasonable proportion of patients retained functional vision.

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